FRED TALKS

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Highlights from my conversation about agentic engineering on Lenny's Podcast

SIMON WILLISON'S WEBLOG: ENTRIES · 02 APR 2026 · SOURCE

Highlights from my conversation about agentic engineering on Lenny's Podcast

I was a guest on Lenny Rachitsky's podcast, in a new episode titled An AI state of the union: We've passed the inflection point, dark factories are coming, and automation timelines. It's available on YouTube, Spotify, and Apple Podcasts. Here are my highlights from our conversation, with relevant links.

An AI state of the union: We've passed the inflection point & dark factories are comingLenny's Podcast

Lenny's Podcast571K subscribers

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The November inflection point

> 4:19—The end result of these two labs throwing everything they had at making their models better at code is that in November we had what I call the inflection point where GPT 5.1 and Claude Opus 4.5 came along.
>
> They were both incrementally better than the previous models, but in a way that crossed a threshold where previously the code would mostly work, but you had to pay very close attention to it. And suddenly we went from that to... almost all of the time it does what you told it to do, which makes all of the difference in the world.
>
> Now you can spin up a coding agent and say, build me a Mac application that does this thing, and you'll get something back which won't just be a buggy pile of rubbish that doesn't do anything.

Software engineers as bellwethers for other information workers

> 5:49—I can churn out 10,000 lines of code in a day. And most of it works. Is that good? Like, how do we get from most of it works to all of it works? There are so many new questions that we're facing, which I think makes us a bellwether for other information workers.
>
> Code is easier than almost every other problem that you pose these agents because code is obviously right or wrong—either it works or it doesn't work. There might be a few subtle hidden bugs, but generally you can tell if the thing actually works.
>
> If it writes you an essay, if it prepares a lawsuit for you, it's so much harder to derive if it's actually done a good job, and to figure out if it got things right or wrong. But it's happening to us as software engineers. It came for us first.
>
> And we're figuring out, OK, what do our careers look like? How do we work as teams when part of what we did that used to take most of the time doesn't take most of the time anymore? What does that look like? And it's going to be very interesting seeing how this rolls out to other information work in the future.

Lawyers are falling for this really badly. The AI hallucination cases database is up to 1,228 cases now!

Plus this bit from the cold open at the start:

> It used to be you'd ask ChatGPT for some code, and it would spit out some code, and you'd have to run it and test it. The coding agents take that step for you now. And an open question for me is how many other knowledge work fields are actually prone to these agent loops?

Writing code on my phone

> 8:19—I write so much of my code on my phone. It's wild. I can get good work done walking the dog along the beach, which is delightful.

I mainly use the Claude iPhone app for this, both with a regular Claude chat session (which can execute code now) or using it to control Claude Code for web.

Responsible vibe coding

> 9:55 If you're vibe coding something for yourself, where the only person who gets hurt if it has bugs is you, go wild. That's completely fine. The moment you ship your vibe coding code for other people to use, where your bugs might actually harm somebody else, that's when you need to take a step back.

See also When is it OK to vibe code?

Dark Factories and StrongDM

> 12:49 The reason it's called the dark factory is there's this idea in factory automation that if your factory is so automated that you don't need any people there, you can turn the lights off. Like the machines can operate in complete darkness if you don't need people on the factory floor. What does that look like for software? [...]
>
> So there's this policy that nobody writes any code: you cannot type code into a computer. And honestly, six months ago, I thought that was crazy. And today, probably 95% of the code that I produce, I didn't type myself. That world is practical already because the latest models are good enough that you can tell them to rename that variable and refactor and add this line there... and they'll just do it—it's faster than you typing on the keyboard yourself.
>
> The next rule though, is nobody *reads* the code. And this is the thing which StrongDM started doing last year.

I wrote a lot more about StrongDM's dark factory explorations back in February.

The bottleneck has moved to testing

> 21:27—It used to be, you'd come up with a spec and you hand it to your engineering team. And three weeks later, if you're lucky, they'd come back with an implementation. And now that maybe takes three hours, depending on how well the coding agents are established for that kind of thing. So now what, right? Now, where else are the bottlenecks?
>
> Anyone who's done any product work knows that your initial ideas are always wrong. What matters is proving them, and testing them.

> We can test things so much faster now because we can build workable prototypes so much quicker. So there's an interesting thing I've been doing in my own work where any feature that I want to design, I'll often prototype three different ways it could work because that takes very little time.

I've always loved prototyping things, and prototyping is even more valuable now.

> 22:40—A UI prototype is free now. ChatGPT and Claude will just build you a very convincing UI for anything that you describe. And that's how you should be working. I think anyone who's doing product design and isn't vibe coding little prototypes is missing out on the most powerful boost that we get in that step.
>
> But then what do you do? Given your three options that you have instead of one option, how do you prove to yourself which one of those is the best? I don't have a confident answer to that. I expect this is where the good old fashioned usability testing comes in.

More on prototyping later on:

> 46:35—Throughout my entire career, my superpower has been prototyping. I've been very quick at knocking out working prototypes of things. I'm the person who can show up at a meeting and say, look, here's how it could work. And that was kind of my unique selling point. And that's gone. Anyone can do what I could do.

This stuff is exhausting

> 26:25—I'm finding that using coding agents well is taking every inch of my 25 years of experience as a software engineer, and it is mentally exhausting. I can fire up four agents in parallel and have them work on four different problems. And by like 11 AM, I am wiped out for the day. [...]
>
> There's a personal skill we have to learn in finding our new limits—what's a responsible way for us not to burn out.
>
> I've talked to a lot of people who are losing sleep because they're like, my coding agents could be doing work for me. I'm just going to stay up an extra half hour and set off a bunch of extra things... and then waking up at four in the morning. That's obviously unsustainable. [...]
>
> There's an element of sort of gambling and addiction to how we're using some of these tools.

Interruptions cost a lot less now

> 45:16—People talk about how important it is not to interrupt your coders. Your coders need to have solid two to four hour blocks of uninterrupted work so they can spin up their mental model and churn out the code. That's changed completely. My programming work, I need two minutes every now and then to prompt my agent about what to do next. And then I can do the other stuff and I can go back. I'm much more interruptible than I used to be.

My ability to estimate software is broken

> 28:19—I've got 25 years of experience in how long it takes to build something. And that's all completely gone—it doesn't work anymore because I can look at a problem and say that this is going to take two weeks, so it's not worth it. And now it's like... maybe it's going to take 20 minutes because the reason it would have taken two weeks was all of the sort of crufty coding things that the AI is now covering for us.
>
> I constantly throw tasks at AI that I don't think it'll be able to do because every now and then it does it. And when it doesn't do it, you learn, right? But when it *does* do something, especially something that the previous models couldn't do, that's actually cutting edge AI research.

And a related anecdote:

> 36:56—A lot of my friends have been talking about how they have this backlog of side projects, right? For the last 10, 15 years, they've got projects they never quite finished. And some of them are like, well, I've done them all now. Last couple of months, I just went through and every evening I'm like, let's take that project and finish it. And they almost feel a sort of sense of loss at the end where they're like, well, okay, my backlog's gone. Now what am I going to build?

It's tough for people in the middle

> 29:29—So ThoughtWorks, the big IT consultancy, did an offsite about a month ago, and they got a whole bunch of engineering VPs in from different companies to talk about this stuff. And one of the interesting theories they came up with is they think this stuff is really good for experienced engineers, like it amplifies their skills. It's really good for new engineers because it solves so many of those onboarding problems. The problem is

> the people in the middle. If you're mid-career, if you haven't made it to sort of super senior engineer yet, but you're not sort of new either, that's the group which is probably in the most trouble right now.

I mentioned Cloudflare hiring 1,000 interns, and Shopify too.

Lenny asked for my advice for people stuck in that middle:

> 31:21—That's a big responsibility you're putting on me there! I think the way forward is to lean into this stuff and figure out how do I help this make me better?
>
> A lot of people worry about skill atrophy: if the AI is doing it for you, you're not learning anything. I think if you're worried about that, you push back at it. You have to be mindful about how you're applying the technology and think, okay, I've been given this thing that can answer any question and *often* gets it right. How can I use this to amplify my own skills, to learn new things, to take on much more ambitious projects? [...]
>
> 33:05—Everything is changing so fast right now. The only universal skill is being able to roll with the changes. That's the thing that we all need.
>
> The term that comes up most in these conversations about how you can be great with AI is *agency*. I think agents have no agency at all. I would argue that the one thing AI can never have is agency because it doesn't have human motivations.
>
> So I'd say that's the thing is to invest in your own agency and invest in how to use this technology to get better at what you do and to do new things.

It's harder to evaluate software

The fact that it's so easy to create software with detailed documentation and robust tests means it's harder to figure out what's a credible project.

> 37:47 Sometimes I'll have an idea for a piece of software, Python library or whatever, and I can knock it out in like an hour and get to a point where it's got documentation and tests and all of those things, and it looks like the kind of software that previously I'd have spent several weeks on—and I can stick it up on GitHub

> And yet... I don't believe in it. And the reason I don't believe in it is that I got to rush through all of those things... I think the quality is probably good, but I haven't spent enough time with it to feel confident in that quality. Most importantly, I *haven't used it yet*.
>
> It turns out when I'm using somebody else's software, the thing I care most about is I want them to have used it for months.
>
> I've got some very cool software that I built that I've *never used*. It was quicker to build it than to actually try and use it!

The misconception that AI tools are easy

> 41:31—Everyone's like, oh, it must be easy. It's just a chat bot. It's not easy. That's one of the great misconceptions in AI is that using these tools effectively is easy. It takes a lot of practice and it takes a lot of trying things that didn't work and trying things that did work.

Coding agents are useful for security research now

> 19:04—In the past sort of three to six months, they've started being credible as security researchers, which is sending shockwaves through the security research industry.

See Thomas Ptacek: Vulnerability Research Is Cooked.

At the same time, open source projects are being bombarded with junk security reports:

> 20:05—There are these people who don't know what they're doing, who are asking ChatGPT to find a security hole and then reporting it to the maintainer. And the report looks good. ChatGPT can produce a very well formatted report of a vulnerability. It's a total waste of time. It's not actually verified as being a real problem.

A good example of the right way to do this is Anthropic's collaboration with Firefox, where Anthropic's security team *verified* every security problem before passing them to Mozilla.

OpenClaw

Of course we had to talk about OpenClaw! Lenny had his running on a Mac Mini.

> 1:29:23—OpenClaw demonstrates that people want a personal digital assistant so much that they are willing to not just overlook the security side of things, but also getting the thing running is not easy. You've got to create API keys and tokens and install stuff. It's not trivial to get set up and hundreds of thousands of people got it set up. [...]
>
> The first line of code for OpenClaw was written on November the 25th. And then in the Super Bowl, there was an ad for AI.com, which was effectively a vaporware white labeled OpenClaw hosting provider. So we went from first line of code in November to Super Bowl ad in what? Three and a half months.

I continue to love Drew Breunig's description of OpenClaw as a digital pet:

> A friend of mine said that OpenClaw is basically a Tamagotchi. It's a digital pet and you buy the Mac Mini as an aquarium.

Journalists are good at dealing with unreliable sources

In talking about my explorations of AI for data journalism through Datasette:

> 1:34:58—You would have thought that AI is a very bad fit for journalism where the whole idea is to find the truth. But the flip side is journalists deal with untrustworthy sources all the time. The art of journalism is you talk to a bunch of people and some of them lie to you and you figure out what's true. So as long as the journalist treats the AI as yet another unreliable source, they're actually better equipped to work with AI than most other professions are.

The pelican benchmark

Of course we talked about pelicans riding bicycles:

> 56:10—There appears to be a very strong correlation between how good their drawing of a pelican riding a bicycle is and how good they are at everything else. And nobody can explain to me why that is. [...]
>
> People kept on asking me, what if labs cheat on the benchmark? And my answer has always been, really, all I want from life is a really good picture of a pelican riding a bicycle. And if I can trick every AI lab in the world into cheating on benchmarks to get it, then that just achieves my goal.

> 59:56—I think something people often miss is that this space is inherently funny. The fact that we have these incredibly expensive, power hungry, supposedly the most advanced computers of all time. And if you ask them to draw a pelican on a bicycle, it looks like a five-year-old drew it. That's really funny to me.

And finally, some good news about parrots

Lenny asked if I had anything else I wanted to leave listeners with to wrap up the show, so I went with the best piece of news in the world right now.

> 1:38:10—There is a rare parrot in New Zealand called the Kākāpō. There are only 250 of these parrots left in the world. They are flightless nocturnal parrots—beautiful green dumpy looking things. And the good news is they're having a fantastic breeding season in 2026,
>
> They only breed when the Rimu trees in New Zealand have a mass fruiting season, and the Rimu trees haven't done that since 2022—so there has not been a single baby kākāpō born in four years.
>
> This year, the Rimu trees are in fruit. The kākāpō are breeding. There have been dozens of new chicks born. It's a really, really good time. It's great news for rare New Zealand parrots and you should look them up because they're delightful.

Everyone should watch the live stream of Rakiura on her nest with two chicks!

YouTube chapters

Here's the full list of chapters Lenny's team defined for the YouTube video:

- 00:00: Introduction to Simon Willison
- 02:40: The November 2025 inflection point
- 08:01: What's possible now with AI coding
- 10:42: Vibe coding vs. agentic engineering
- 13:57: The dark-factory pattern
- 20:41: Where bottlenecks have shifted

- 23:36: Where human brains will continue to be valuable
- 25:32: Defending of software engineers
- 29:12: Why experienced engineers get better results
- 30:48: Advice for avoiding the permanent underclass
- 33:52: Leaning into AI to amplify your skills
- 35:12: Why Simon says he’s working harder than ever
- 37:23: The market for pre-2022 human-written code
- 40:01: Prediction: 50% of engineers writing 95% AI code by the end of 2026
- 44:34: The impact of cheap code
- 48:27: Simon’s AI stack
- 54:08: Using AI for research
- 55:12: The pelican-riding-a-bicycle benchmark
- 59:01: The inherent ridiculousness of AI
- 1:00:52: Hoarding things you know how to do
- 1:08:21: Red/green TDD pattern for better AI code
- 1:14:43: Starting projects with good templates
- 1:16:31: The lethal trifecta and prompt injection
- 1:21:53: Why 97% effectiveness is a failing grade
- 1:25:19: The normalization of deviance
- 1:28:32: OpenClaw: the security nightmare everyone is looking past
- 1:34:22: What’s next for Simon
- 1:36:47: Zero-deliverable consulting

- 1:38:05: Good news about Kakapo parrots

The lies I used to tell myself

CATE HALL · 23 JAN 2026 · SOURCE

1. What doesn't kill you makes you stronger.

When I was 31, my husband of four years, who I loved more than my own life, left me completely out of the blue. We'd had an incredible, epic romance — drawn together like magnets, we married on our third date and spent the years that followed burrowing ever closer together. Maybe that was the problem — I can't say for sure, because I never got a satisfying explanation out of him. One day I was the protagonist of a Princess Bride-worthy love story; 48 hours later, he moved out.

From the time we met, I had never contemplated the possibility of living without him, and I didn't want to. I wanted to die for a long time — at least a couple of years. But as I slowly learned to sequester away thoughts of him and the emotions they brought up, I became convinced of a silver lining: The experience would make me impenetrable. If I survived it, I would know I could survive anything, and nothing would be able to hurt me again.

It took the better part of a decade and another deeply identity-shifting relationship, with the man who became my second husband, for me to see that I had it all wrong: Impenetrability means avoiding the parts of life that are most life-like. There's a reason the visual metaphor for love is an arrow.

The near-fatal catastrophes of our lives don't always make us stronger. Surviving one hard thing does provide evidence that you can survive other hard things, which is easy to *mistake* for getting stronger. But in reality they often just make us brittle, make us dissociate and flee from the situations that remind us we can be broken.

2. When you know, you know.

Mario Gabriele of the Generalist had a great "50 Things" style post a few months ago that I found myself nodding along to almost the whole way through. The glaring exception was Number 33: "Love is easy. The difference between bad relationships and meeting your spouse is not a matter of degree, but category. Compatibility is unignorable and effortless."

I believed this completely, once. I've since come to believe that what's unignorable and effortless is chemistry. Chemistry can be powerful enough to turn your life sideways, but it's not the same thing as compatibility. Compatibility is litigated on the timescale of years and decades; it's not a momentary feeling but a description of what happens during an extended course of change. Do you grow toward one another, or do you grow apart? Does the relationship help you move closer to the person you aspire to be?

I "just knew" with my first husband. I wasn't sure with my second. Four years in, the signs have flipped.

3. If it were a good idea, someone would be doing it already.

There's a belief, especially common among wonky types, that you don't find $20 bills on the sidewalk — if an opportunity were real, someone would have seized it already. This sounds worldly and wise, but it's actually a form of defensive thinking. It assumes the current way is best, that everything worth doing is being done.

In my experience, this is just false. When I started playing poker, I discovered that physical tells were basically a wide-open field — despite the fact that most pros believed the topic was played out. If you think you see an opportunity others have missed, you should have a story for why they're missing it, and you should ask around to stress-test that story. But if no one can give you a satisfying reason, don't assume one exists. Sometimes the answer really is just inertia or fear.

4. If it's worth doing, it's worth doing well.

Almost nothing is worth doing "well." Many things can and should be done to a minimal standard of quality, because anything more is a waste of effort that can be better allocated. The exceptions aren't things that should be done "well," but things that should be done to an exceptionally high standard, probably a much higher one than you see modeled by the people around you.

This is not semantics; most people really "do everything the way they do anything" — whether that's in a low-effort or high-effort way. But the point of phoning it in on a lot of stuff is to save your energy to expend on the 1 in 1000 things that really matter … and then to actually spend it.

A recent example: I decided that I want to narrate my own audiobook for *You Can Just Do Things*. When I told a friend in the industry this and asked whether they knew any good coaches to work with, they said that hiring a coach wasn't normal or necessary, because my publisher would decide based on a sample whether I was good enough at reading to do it, and if so give me a producer to work with day-of. But it "1 in 1000" matters to me whether the audiobook is so good no one would ever think of returning it on account of the narration, so I'm not interested in the normal-shaped solution that constitutes doing it "well," I'm interested in the one that causes me to come to mind when someone asks this question. In this case, that means hiring multiple coaches and practicing for months.

5. If you like your job, you're doing something wrong.

Back when I was more alienated from my emotions, I thought about work like this: You have a budget of energy to spend, and the goal is to efficiently convert that energy into impact. You identify the activity that generates the highest impact per unit of energy, and then you spam that button for as many waking hours as you can stomach. Is it any wonder I was always burning out?

What I failed to understand is that energy is not fixed. It's dramatically affected by what you choose to do — some activities increase your energy over time, others deplete it. This feels so obvious to me now that I feel kind of stupid including it here, but I know many people (they are endemic to the Bay Area) who still labor under the false belief. Whether you enjoy what you're doing affects your ability to do it year in and year out, and dismissing this as "selfish" doesn't make it untrue. As my husband puts it, the best use of effortful motivation is to engineer your life to require less of it.

6. “All people are insane. They will do anything at any time, and God help anybody who looks for reasons.”

This line, from Kurt Vonnegut, used to neatly encapsulate the way I understood other people — which is to say, not at all. Trying to figure out why people did what they did, or imagine what they’d do next, seemed like a fool’s errand. I was judgmental about it, too: I assumed this was a problem, not with me, but with everyone else.

I didn’t fundamentally move beyond this belief until recent years, when I came into contact with the Enneagram. I imagine there are other personality typing systems that can do something similar, but the Enneagram was magical for me because it clued me into the fact that people can have very different motivational systems from my own that are still internally coherent and produce good things in the world.

People’s behavior was previously incomprehensible to me because I was imputing my own motivational system to them, trying to figure out what would cause me to act the way they did — essentially, concluding that *their* actions made no sense in light of *my* goals. But people have wildly diverse emotional hardware, which determines what’s rational for them. And my particular personality — independent, moralistic, systematic, challenge-seeking, competitive — is unusual, so I was especially poorly placed to measure others by the yardstick of myself.

This shift in perspective was dramatically good for me and for my relationships with other people. It let me see the ecosystem of psychologies as a kind of miracle — wow, it really does take all kinds — rather than a prompt for cosmic horror. And it gave me confidence that, if I invested in understanding people more deeply, they would eventually cease to be total mysteries, prone to bizarre and unreasonable behavior. As far as I can tell, this is the better part of emotional intelligence.

7. Money can’t buy happiness.

The good version of this advice is: You’ll still have to work at being happy, even if you’re rich. Money in your bank account won’t do it for you, and wealth comes with its own pathologies, like endless status-chasing. But the common gloss — that money stops mattering past some modest threshold — is basically false.

You may have heard there’s research showing money doesn’t buy happiness. That research was explosively popular precisely because it was counterintuitive — it betrayed the commonsense intuition that more money means more freedom, more options, more ability to help people you care about. Turns out it was also riddled with methodological problems. More recent work by Matthew Killingsworth (nominative determinism FTW), using large-scale studies that pinged participants about their happiness at random moments, found steady returns to well-being with

rising income, no plateau in sight. The myth persists because it serves a psychological function: It helps people without money feel better about their situation, and helps the wealthy downplay their advantages. But it's not true.

8. Intuition is fake and rationality solves everything.

I used to be horribly judgmental of people who made decisions based on intuition, as in, "I just do what feels right." And then I noticed that whenever I overrode my intuitions, I made *terrible* decisions. I hired the wrong people, started projects with the wrong people, dated the wrong people, ate food that made me feel bad, did forms of exercise I found unfun and injurious. I don't have a satisfying theory for why this is, but I know what happens: When I shift from emotional intuition to a logical story about a decision, it's much easier for my reasoning to get hijacked by plausible-sounding directives that turn out to be nonsense.

It took me a long time to learn this, and that might have something to do with my training in law and poker. Overriding intuition in favor of explicit reasoning makes sense in contexts that reward systematic rationality, where you will get separated from your money if you count on gut feelings. But most of life isn't a closed system with enumerated rules and clear outcomes. If you try to explicitly name all the factors that make someone a good friend or co-founder, you might come up with some good indicators, but your crude map will also mislead you about the territory.

This is why "wisdom is knowledge that can't be transmitted." Wise people have navigational skill, not a long list of rules.

Sign up to be notified when my book, You Can Just Do Things, is available for purchase.

Ownership

THORSTEN BALL · 08 JUL 2026 · SOURCE

Thoughts on ownership I sent to the Amp team in internal note

The following is an internal Slack message I sent to my Amp teammates after I had a conversation with one of them about ownership. It's only lightly edited.

I shared it before, but not here, because I didn't think too much of it. Then today, someone said their CTO shared my post with them and I thought: well, now I have to put it in the newsletter, don't I? So here we are.

Below, after the Slack post, I added some thoughts on juniors on how I see this advice applying to them.

Just had a (great) conversation about ownership and engineering here and I realized that I often use the phrase "ownership" or allude to it, but haven't explained what "ownership" means to me in a while.

So, ownership.

If I ask you "can you own this?" or "can you take care of this?" or "are you on it?" — what I'm doing is I'm asking you to *own* it, to own the solution of a problem from end to end. From "we have a problem" to "we don't have to think about it again."

That means, when you say that you're owning something, the expectation is that you…

- Think about what the problem actually is. Maybe you already have a solution in mind, without having thought about what we're actually trying to solve here. Maybe you think "the problem is that we need to migrate from using X to using Y", but that's not a problem, that's a solution. The problem is likely something like "performance is bad", "it's not stable", "it fails for customer x". Maybe there's other possible solutions to that? Think about those. What are the tradeoffs? What's the best solution to go with considering these tradeoffs?

- Think about edge cases. What are they? Which ones are important? Which ones can we ignore?

- Think about failures. Network failures are a given, for example. How do we handle them? Retry? Well, how often? How long?

- Think about data flow. How much data is involved here? Does data need to be migrated? Cleaned up? How can I get my hands on data to properly test this? What invariants are in the data? What assumptions do I have about the shape of the data that I haven't confirmed yet?

- Think about how you'd test this. How can I know that what I built is correct or not? Are tests enough? Do I need to manually poke at things? Is the difference visible on a screenshot or in a video?

- How would we announce this? How do we communicate it? Can you picture it? How does it fit into the larger picture of our roadmap? Questions or concerns in that area — push back! ask!

- Do the work, with precision, with care, with a sense of urgency, with calmness. Do not half-ass things. Before you merge, ask yourself: am I proud of this? would I show this to John Carmack and say “here’s what I built, under these constraints, with these tradeoffs?”

- Test it manually. Yes, there’s automated tests. But in 99% of cases you can manually test or confirm that what you built works: you can run it yourself, you can ask an agent to run through test scenarios, you can poke at the data before and after, you can take screenshots, you can make a demo. Are you sure that what you did actually solves the problem?

- Make sure it lands in production and *works in production*. Is it deployed? Did the deploy fail? Do you need to activate a feature flag? Does the feature flag work? Can you use it in production? Can you confirm it’s actually deployed?

- If you think your colleagues needs to know about this change, because it’s new feature they should all test, or it’s a new convention in codebase, or maybe it’s a tricky thing everybody needs to be aware of, or something else: let them know! Do not underestimate peripheral vision: knowing that person X yesterday changed the behavior of how Z works might save person Y three hours of debugging today when a bug report related to Z comes in.

- Do customers need to know? Who reported the bug? Who’s blocked? Let them know.

- Does the world need to know? Announce that it’s out.

- Are there follow-ups? Do you need to check on what you shipped in the logs? A week later maybe?

Yes, that’s a lot. And there’s actually more, because I’m sure I forgot some stuff.

But that’s how you build a product in a small team. We don’t have PMs, we don’t have a Q&A department. We’re small, but we’re *great*, we can do all of that.

And it’s always okay to ask for help, it’s okay to ask questions, it’s okay to redo things and triple-check. What’s not okay is to implicitly assume that someone else will do the things here that you haven’t thought about.

“How does this apply juniors? You can’t expect them to really do all of that?”

I’ve been asked these questions, or variations of them, after I shared the thoughts above and here’s my answer.

I do *not* expect juniors to *do* all of these things right away. But I would expect them to read through the list and *aspire* to one day be able to do all of these things. Until then, they can and should ask for help.

In fact, I don’t expect *anybody* to *always* do *all of these things* for *everything*. It’s a mental checklist of things to consider — problem, edge cases, tradeoffs, deployment, customers, messaging, … — but for quite a few things there aren’t edge cases to consider. Or big tradeoffs to weigh. Or deployment is a solved problem. And maybe someone else does the messaging for you.

And I imagine that most of these things you shouldn’t even consider when you work in a company with, say, 5000 employees. I’ve never worked in a company that large, only startups, so I can’t speak to how to successfully ship a software feature at Apple, end to end.

But when you work in a small company in which there’s only a single department, when you want to build things you’re proud of, when you work with me and you say own something, I expect you to keep these things in mind and run through them before you declare something as done.

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